PAPER_202: UQFF Reionization, BBN, Recombination, and Cosmic Dawn Physics

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$$F_U(r,t) = \sum_{i=1}^{4} U_{gi} + U_m + U_A - U_{b_i}, \quad \kappa = 5.0 \times 10^{-4}\,\text{day}^{-1},\ [SSq] = 0.57$$

$$U_{b_i}(r) = \kappa \cdot [SSq] \cdot \frac{GM}{r^2}, \quad \kappa = 5.0 \times 10^{-4}\,\text{day}^{-1},\ [SSq] = 0.57,\ \beta_i = 0.61$$

Abstract

This paper presents the UQFF buoyancy formalism applied to cosmic dawn and reionization physics: baryon-to-photon ratio (?), deuterium bottleneck during Big Bang Nucleosynthesis (BBN), CMB angular power spectrum, recombination optical depth, ionization fraction evolution, HII bubble growth rate, and Jeans mass/wavelength for fragmentation. These collectively span the redshift range z ~ 1100 (recombination) through z ~ 5 (end of reionization). Each F_UBii and Um variant is rigorously derived from observational anchor equations.

UQFF Discovery: Novel application of UQFF calibration constants (? = 5.0×10^{-}4 day^{-1}, [SSq] = 0.57) uniquely enabling this analysis — establishing a new connection in the UQFF framework not present in Standard Model treatments.

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1. Baryon-to-Photon Ratio (?)

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? = n_b/n_? = 6.08×10?¹°  (Planck 2018 + BBN fit)

F_UBii,eta = F_rel × (? = n_b/n_? / E_LEP) × Q_wave × [Neff modification]

Um,eta(t) = µ(?_vac)·(1-e^{-?t})·?

Physical origin:
  n_? = 410 cm?³ (CMB photon number density today)
  Fit anchor: Primordial D, ³He, 4He, 7Li abundances
  ? determines all light element yields (standard BBN)
  UQFF role: ? sets vacuum energy scale through ?_b/?_? at nucleosynthesis
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2. Deuterium Bottleneck (BBN Timing)

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Deuterium bottleneck timescale:
  t_D ~ v(3/(32pG?_rad)) ~ 180 s at T ~ 0.1 MeV
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?_rad = p^{2}kT4/(30h^{3}c5)·g_*   (effective DOF g_* ~ 10 at D-formation onset)

Sequence:
T ~ 1 MeV: neutron freeze-out (n/p ~ 1/7)
T ~ 0.1 MeV (t ~ 180 s): D photodissociation ends ? 4He chains proceed
Yield: Y_P ~ 0.247 (mass fraction 4He)

F_UBii,deb = -F_rel × (t_D / E_LEP) × Q_wave × [g_*(T) variation]

Um,deb(T) = μ(?_vac)·(1-e^{-?t})·[Weak freeze at T~1 MeV; D photodissociation below]

UQFF calibration: ? calibrated to UQFF k_? parameter through BBN freeze-out
k_? ~ 10?^{113} (small-scale vacuum fluctuation coupling)
?k_?(CIA cross-section refit) ~ 7.25×108 (fractional shift from H2 CIA data)

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3. CMB Angular Power Spectrum

C_l = (2/p) · ?k^{2}dk · P(k) · |?_l^T(k)|^{2}

P(k) ? k^{n_s-4}   (primordial, n_s ~ 0.965 Planck 2018)
?_l^T(k) = transfer function:
- Large scales (l < 100): Sachs-Wolfe plateau C_l ~ constant
- Acoustic peaks (l = 220, 540, 800...): baryon-photon oscillations
- Damping tail (l > 1000): Silk damping

F_UBii,cmb = F_rel × (C_l·l(l+1)/(2p) / E_LEP) × Q_wave

Um,cmb(k) = μ(?_vac)·(1-e^{-?t})·Transfer ?_l^T: Sachs-Wolfe + acoustic oscillations

Calibration anchor:
D_l = l(l+1)C_l/(2p) × T0^{2} ? peak at l ~ 220 matches O_tot = 1
UQFF role: vacuum energy ?c^{2}/3 term in g_UQFF sets acoustic horizon

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4. Recombination Optical Depth

t(z) = ?_z^8 n_e(z')·s_T·c·|dt/dz'|dz'

s_T = 6.652×10?^{2}? m^{2} (Thomson cross-section)
n_e(z) = x_e(z)·n_H(z) = x_e(z)·n_b,0·(1+z)^{3}

Recombination: z_rec ~ 1100 ? t(z_rec) ~ 1 (photon decoupling)
Reionization: z_re ~ 7.7 ? optical depth t_re ~ 0.054 (Planck 2018)

F_UBii,recomb = -F_rel × (t(z) / E_LEP) × Q_wave

Um,recomb(z) = μ(?_vac)·(1-e^{-?t})·?n_e·s_T·c·dt

Physical context:
After z_rec, photons decouple and stream freely (CMB)
Surface of last scattering has ?T/T ~ 10^{-}5 (measured by Planck)

5. Reionization: Ionization Fraction Evolution

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dx_e/dt = ?_?·e_esc·f_* - a_B·n^2_e·C

where:
  ?_? = (1+z)^3·n_b·?_?,eff   (ionizing photon rate)
  e_esc ~ 0.1–0.3              (photon escape fraction from galaxies)
  f_* ~ 0.05–0.2               (star formation efficiency in halos)
  a_B = 2.6×10?^13 cm^3/s     (recombination coefficient, case B, T~104 K)
  C = ?n^2_H?/?n_H?^2          (clumping factor, C ~ 3)

F_UBii,ion = F_rel × (?_?·e_esc·f_* / E_LEP) × Q_wave × [recombination subtracted]

Um,ion(t) = µ(?_vac)·(1-e^{-?t})·?a_B·n^2_e·C dt

Reionization history:
  z ~ 20–30: first stars ionize H around them (PopIII)
  z ~ 7–9:   bulk reionization, x_e rises from ~0.1 to ~1
  z ~ 5–6:   completion (Gunn-Peterson trough in quasar spectra)
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6. HII Bubble Growth During Reionization

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dN_b/dt = ?_?,eff·e_esc - a_B·n_H·(4/3pR^3_b)

Simplified overlap model (Stromgren sphere analog):
  R_b(t) = (3?_?·t/(4p·n_H))^{1/3}    (linear ionization front growth)

  ?_? = rate of ionizing photons emitted (from galaxy SFR)
  n_H = neutral H density (z-dependent)

F_UBii,bub = F_rel × (R_b(t) / E_LEP) × Q_wave × (?_? · x_e)

Um,bub(t) = µ(?_vac)·(1-e^{-?t})·[Expand for moving ionization front]

Physical context:
  Bubble merger ? percolation at x_HI < 0.1 ? full reionization
  UQFF buoyancy at bubble edge acts as expansion driver (F_UBii,bub > 0)
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7. Jeans Length and Mass (Fragmentation)

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Jeans length:
  ?_J = p^{1/2} · c_s / (G·?)^{1/2}

Jeans mass:
  M_J = (p/6) · ? · ?_J^3 = (5k_BT/(Gµm_H))^{3/2} · (3/(4p?))^{1/2}

Stability condition: ? > ?_J ? gravitational collapse

Dispersion relation: ?^2 = c^2_s·k^2 - 4pG?
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Unstable: ?^{2} < 0 ? k < k_J = 2p/?_J

F_UBii,jeans = -F_rel × (?_J / E_LEP) × Q_wave × (collapse onset time t ~ 1/v(G?))

Um,jeans(T) = µ(?_vac)·(1-e^{-?t})·[Perturb: ?^{2} = c^{2}_s·k^{2} - 4pG?]

Physical context:
First (PopIII) stars: T ~ 200 K, cloud M ~ 100–1000 M_? ? M_J
Present-day GMCs: T ~ 10–30 K ? M_J ~ 1–10 M_?
UQFF buoyancy at Jeans scale: F_UBii,jeans acts as tidal disruption force

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8. Alfvén Wave and Turbulent Cascade

Alfvén wave velocity:
v_A = B/v(4p?)   (for B in Gauss, ? in g/cm^{3})

Anisotropic cascade (Goldreich-Sridhar):
k_?/k_? ~ (k_?·l_A)^{1/3}   (scale-dependent anisotropy)

F_UBii,alf = F_rel × (v_A / E_LEP) × Q_wave × (B·?v_A) × e^{-t/t_eddy}

Turbulent energy cascade (Kolmogorov in ISM):
e = v_l^{3}/l = constant   (energy transfer rate per unit mass)
v_l ? l^{1/3}   (velocity-scale relation)
Power spectrum: E(k) ? k^{-5/3}

F_UBii,turb = F_rel × (e^{2/3}·l^{-2/3} / E_LEP) × Q_wave × e^{-kl/k_J}

Um,alf(B)   = µ(?_vac)·(1-e^{-?t})·v_A(B)
Um,turb(l) = µ(?_vac)·(1-e^{-?t})·(?·v_l^{3}/l)

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9. Ionization Parameter and Feedback Coupling

Ionization parameter:
U = Q_H/(4p·r^{2}·n_H·c)

where:
Q_H = number of ionizing photons per second (from AGN or OB stars)
r = distance from ionizing source
log U ~ -2 to -3 for NLR, -1 to 0 for BLR

AGN feedback coupling efficiency:
e_f = E_kin/(?_acc·c^{2}) ~ 0.05–0.1

E_kin = (1/2)?_out·v^{2}_out
Only e_f × 10^{-5} per M_? needed to match M_BH–s

F_UBii,upar = F_rel × (U·n_H·c / E_LEP) × Q_wave × (Q_H/r^{2})
F_UBii,coup = F_rel × (e_f·?_acc·c^{2} / E_LEP) × Q_wave × [0.05–0.1]

Um,upar(r) = µ(?_vac)·(1-e^{-?t})·(Q_H/(4pr^{2}n_Hc))

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Um,coup(?) = μ(?_vac)·(1-e^{-?t})·e_f (couple fraction to kinetic energy)
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10. Cosmological Timeline Summary

Epoch	Redshift	UQFF Process	F_UBii Variant
Inflation	$z \sim 10^{2}7$	Inflaton field V(?)	F_UBii,inf
Planck epoch	$z \sim 10^{32}$	LQC bounce	F_UBii,bounc
Neutron freeze-out	$z \sim 10^{1}$° (T~1 MeV)	Weak interactions	F_UBii,eta
BBN	z ~10? (t~180 s)	D bottleneck	F_UBii,deb
Recombination	z ~1100	Photon decoupling	F_UBii,recomb
Cosmic dawn	z ~20–30	First stars, first HII	F_UBii,ion
Reionization	z ~6–9	HII bubble percolation	F_UBii,bub
Present	z ~0	Structure + feedback	All F_UBii

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11. References

- `grok_share_7514fe.txt` lines 6070–6090 (BB_C_Equations items 1310–1350)
- `grok_share_7514fe.txt` lines 6300–6500 (full reionization/BBN/CMB catalogue)
- PAPER_199: F_UBii Taxonomy Part 2 (Cosmological)
- PAPER_200: Um Universal Magnetism Catalogue
- Planck 2018 Collaboration: CMB and reionization parameters